Perceptual and Acoustic Correlates of Racial Identity in Text-to-Speech Voices

Anonymous submission to Interspeech 2026

Abstract

Voice-activated artificial intelligence (Voice-AI) systems are increasingly human-like in production, raising questions regarding how these synthetic voices are socially perceived. In this study, we show that listeners are generally capable of perceiving a sense of speaker race when hearing synthesized speech, and that they even engage in racial stereotyping. We do, however, see a bias towards rating voices as sounding 'White'. To explore this, we train a Gradient Boosted Decision Tree classifier on acoustic features to distinguish between voices rated as Black from White. Our model performs with an accuracy of 65%. From this analysis, we find that voices with lower Residual H1* and spectral tilt in the lower/higher frequencies are more regularly rated as Black. Noise measurements, as well as formants (F1, F2, and F4), also seem to vary reliably between race labels. These results have implications regarding the development, use, and efficacy of racially-marked AI [1].

Index Terms: Text-to-Speech, Acoustic Phonetics

1. Introduction

Research has shown that listeners can ascertain information about any given individual's personality and/or their emotional state, just from the sound of their voice [1, 2, 3, 4]. This phenomenon applies to racial perception as well. For example, American English listeners have been shown to reliably distinguish between White and Black Americans, based on phonetic features associated with Mainstream American English (MAE) and African American English (AAE), even in the absence of syntactic or morphosyntactic cues [5, 6]. In addition, the perception of race in voice interacts with the perception of the individual's personality and/or social status. This has been shown repeatedly with American English listeners, who tend to view voices they hear as 'sounding Black' more negatively in different social or emotional contexts [5, 6].

With recent advances in Voice-AI, synthetic speech is increasingly capable of sounding human-like in their production. As speech technology continues to be deployed in high-stakes contexts such as education, healthcare, customer service, and assistive technologies, understanding the social attributes encoded in these voices take on practical and ethical value.

Along these lines, studies have attempted to encode race in synthetic TTS speech [7, 8], though the efficacy of this remains questionable. While some studies working with American English speakers report that listeners converge on race labels when hearing TTS voices [9], others find that Black-sounding synthetic voices are difficult to identify, and that there is a bias towards rating TTS voices as 'White' [7].

Following methodology set forth in [9], the current study aims to identify whether perception of race is possible in fully synthesized speech, particularly when listeners are aware that they are listening to synthetic speech. We also test whether perception of race interacts with personality assessment of any given voice. We focus on the distinction between White American-sounding voices and Black American-sounding voices, as the notion of racial perception and its interaction with personality assessment for these populations in human speech has been well documented. Given that we are working with fully synthetic speech, we also aim to test the impact that so-called 'human-likeness' has on perception of identity with synthetic speech, as human-likeness in synthetic voices promotes anthropomorphism and stronger personality attribution [10, 11].

In this study, we also aim to quantify the acoustic differences that exist between synthetic voices labeled as Black versus White by our listeners. This is important, as differences in the acoustic signal drive racial categorization in human speech. With this in mind, it is well documented that AAE has a different vowel system than MAE [12, 13, 14]. Specifically, the vowels /ɛ, æ, ɪ/ undergo raising relative to MAE [14]. The monophthongization of /eɪ/ and /aɪ/ [15], the fronting of /ɑ/ [16], and the centralization of /i/ [12] relative to MAE have also been documented. We target these vowels specifically in our study. Much less work has examined vowel-independent acoustic features differentiating AAE from MAE. However, existing research suggests that AAE speakers exhibit greater noise in their acoustic signal [5], indicating that AAE speakers may produce speech with a breathier register than MAE speakers. For these reasons, we make an effort to compare both vowel quality and voice quality measures across racially-labeled synthetic voices.

For this experiment, we synthesize voices using an online platform called EasyPeasy AI. We use this platform because they provide a web-based interface with a large variety of available voices to be used for TTS synthesis. Consent to utilize the voices for the purposes of this study was provided by EasyPeasy AI, though explicit information about underlying model structure and training data was not given.

2. Methods

2.1. Experiment

The experiment was hosted on Github pages using a custom web-based interface developed using HTML, CSS, and JavaScript. 150 American English speakers without speech and hearing disorders were recruited using Prolific. Data from six

[1] To ensure author anonymity, the link to the Github repository that contains the experimental materials, as well as the code for the analysis will be added after the review process.

participants were dropped due to them being unable to complete the entire study, leaving 144 total participants. An ethnically representative sample was chosen, with 54% of participants in our study self-identifying as White, 11.8% as Black, 6.94% as Asian, 18.1% as Hispanic/Latino American, and 9% as other. In addition, 56.6% of the participants identified as male, and 43.1% as female. Participants were also distributed across age groups: 6.3% were aged 18–24, 36.1% were aged 25–34, 27.8% were aged 35–44, 18.8% were aged 45–54, and 11.1% were aged 55 or older.

Voices were synthesized on August 20th, 2025. For this study, we focused on English, and utilized two of the available English 'Accents' on the platform: American, which refers to Mainstream American English, and African American, referring to African American English. For each voice, the platform assigns a picture of what the voice might look like if belonging to a real person and a short qualitative description of the voice. Gender (Male/Female) was also labeled by the platform. A voice was considered by the authors as being labeled Black if it was found in the African American tab on the platform, and had a image of a Black person in its description[2]. Likewise, a voice was considered as being labeled as White if it fell into the American tab and was given an image of a White person. Images were not not shown to our participants during the experiment. The gender of the voice was taken directly from the qualitative description. From here on out, we refer to the Black/White and Male/Female labels originally assigned to the voices as the platform-assigned labels. Of the total number of viable voices labeled as Black or White, 18 were filtered out by the first author due to it consistently producing output with poor audio quality.

After voices were aggregated based on their labels, a total of 32 voices were randomly selected, organized by their platform-assigned label: eight labeled as Black Males, eight labeled as Black Females, eight labeled as White Males, and eight labeled as White Females. The number eight was chosen as that was the maximum number of Black Males on the platform at the time of random sampling.

The experiment was divided into two main blocks: Personality and Race. Upon agreeing to the study, participants were informed that they would be hearing synthetic (non-human) speech. Each participant first completed the personality block where they heard 16 randomly selected voices (balanced by race/gender) reading the first six sentences of the Rainbow Passage [17]. For each trial, participants were prompted to rate the voice they heard on six personality characteristics, using 7-point likert scales: friendliness, professionalism, competence, trustworthiness, pleasantness, and funniness [9]. Participants were also asked to rate the human-likeness of the voice on a 7-point scale.

Participants were then prompted to complete the Race block, which contained the 16 voices not used in the Personality block. Each trial consisted of a voice saying one of four sentences, each targeting specific vowels. Sentences avoided phonological features associated with AAE such as fortition, r-deletion, NG-fronting, or syllable-final nasal/liquid deletion, ensuring that the cues listeners may use to differentiate race only come from voice/vowel quality. For each trial, participants were asked about the voice's race, age, and region in the form of multiple choice questions. Race was of particular interest for this study, with other categories included to disguise our exact hypothesis. The categorical options presented to listeners for race judgments were Asian, Black, Hispanic/Latino, or White. The gender of the voice was also assessed for each trial, being put on a 7-point likert scale from Feminine-sounding to Masculine-sounding. There were 64 trials total.

In both blocks, participants were only presented with audio, with no visual cues that could encode voice identity. All audio was amplitude normalized to 72 dB using a Praat script.

Upon completion of the experiment, participants were prompted to answer a series of demographic questions pertaining to their own race, region, gender, age, and exposure to AI technology.

2.2. Acoustic variables

To define a theoretically grounded set of acoustic variables, we employed the Psychoacoustic Model of the Voice (Kreiman et al., 2014; 2021), which characterizes voice quality along perceptually motivated temporal and spectral dimensions that can be parameterized acoustically. We choose this feature set in order to maintain interpretability of our modeling results.

Following a protocol laid out in [18], we extracted fundamental frequency (F0), the first four formant frequencies (F1–F4), and formant dispersion (DF), defined as the mean spacing between adjacent formants in hertz. Harmonic source measures were also extracted, including Residual H1* [3], H2*–H4*, H4*–H2kHz*, and H2kHz*–H5kHz*, all of which were formant-corrected. We also extract measurements related to inharmonic source characteristics: Cepstral Peak Prominence (CPP), Subharmonic-to-Harmonic ratio (SHR), and Energy.

Acoustic variables were extracted at 1-ms frames using VoiceSauce [20] for each vowel token. Frame-level outliers were identified and excluded using a median absolute deviation (MAD) procedure [21], with a threshold of k = 3.5. All frames for each acoustic variables were then normalized by the minimum and maximum value for that feature within gender.

Next, each vowel token was divided into three equal temporal segments relative to its total duration, corresponding to vowel onset, midpoint, and offset. For each vowel token, we computed the mean and the Coefficient of Variability (CoV) of all the acoustic variables at onset, midpoint, and offset. For example, a vowel with a duration of 150 ms would be segmented into three consecutive 50-ms intervals, and mean/CoV would be calculated at each interval, across all acoustic variables. CoV is defined as the standard deviation divided by the mean and has been shown to effectively capture perceptually relevant variability in acoustic measures [22, 23, 24, 18].

Measurements were then centered relative to their phonation modality (i.e., breathy, creaky, breathy-creaky) with modality-specific means calculated separately for each vowel. Modality was assessed by the first author during segmentation, based on visual examination of the spectrogram and waveform. This approach was intended to reduce acoustic variance attributable to phonation types. The normalized/centered mean and CoV values for each acoustic variable at vowel onset, midpoint, and offset across tokens were used as features in the classification model (see §2.3).

[2] Since the image that the platform assigned helped the authors determine the platform-assigned race, a small norming study (n=11) was done to confirm whether the photos were actually representative of the race they were denoting. Results show 100% agreement with the platform.

[3] We use Residual H1* which is the amplitude of the first harmonic, normalized by root mean square energy, rather than the more common spectral tilt measure H1-H2, as it is better correlated with glottal state [19].

2.3. Gradient Boosted Decision Tree learning

Gradient boosting is a supervised machine learning technique that builds a predictive model by combining a series of weak learners. Each weak learner is trained to minimize a loss function, gradually improving performance. The model used in this study was used to distinguish between voices rated as Black and voices rated as White. To analyze our data, we use the R package XGBoost [25].

Optimal model hyper-parameters and number of boosting rounds were chosen by conducting a full grid search using five-fold cross-validation (CV). CV was repeated 50 times per model. CV folds were defined at the speaker level, such that all tokens produced by a given speaker were assigned to the same fold. This ensures that any speaker who was used in a given training fold did not appear in a corresponding test fold. The hyper-parameter values that resulted in the highest average accuracy on test folds were chosen for subsequent testing. The final model used 50 boosting rounds, a maximum tree depth of 4, a learning rate of $\eta = 0.1$, a minimum child weight of 1, and a gamma of $\gamma = 0$. A model with the tuned hyper-parameters and iterations was fit to the entirety of the data and used for calculation of feature importance (see §3.3).

3. Results

3.1. Perceptually-assigned Race labels

Agreement between platform-assigned gender labels and participant ratings was high (Male: 98.6%; Female: 93.6%), collapsed across race. Accordingly, platform-assigned gender was treated as the 'true' gender of the voice in all subsequent analyses.

Perceptually grounded race labels (e.g., 'true' race labels), rather than platform-assigned labels, were derived via k-means clustering, run on %Reported Black (i.e., the % of time any particular voice was rated as 'Black') (RB). RB values were standardized prior to clustering, and a three-cluster solution was selected. Based on the resulting distributions on RB, clusters were interpreted as voices consistently identified as White ($\sim$0–20%), Black ($\sim$50–100%), or Ambiguous ($\sim$20–50%). In other words, race categories for our voices were relabeled based on their cluster assignment. Note that the distribution of the cluster denoting Black-sounding voices is larger than the cluster denoting White-sounding voices, indicating more rater-variability for Black-sounding voices, and suggesting a general bias towards rating voices as White.

Importantly, a majority (5/9) of the voices categorized as Ambiguous were labeled by the platform as 'Black Female', indicating that people had a harder time identifying Black Female voices, as seen with human voices [6].

3.2. Personality Results

Figure 1 shows the ratings for each trait across the perceptually-assigned race categories for voices across both gender labels.

Per-trait linear mixed-effects models were fit with fixed effects for perceptually assigned race, participant ethnicity, and human-likeness, and random intercepts for participant and voice. Participant ethnicity included White, Black, Asian, Hispanic/Latino, and Multiracial American categories. Separate models were fit for perceived female- and male-sounding voices.

Our results show that, for female-sounding voices, perceptually assigned race was not a significant predictor of any trait.

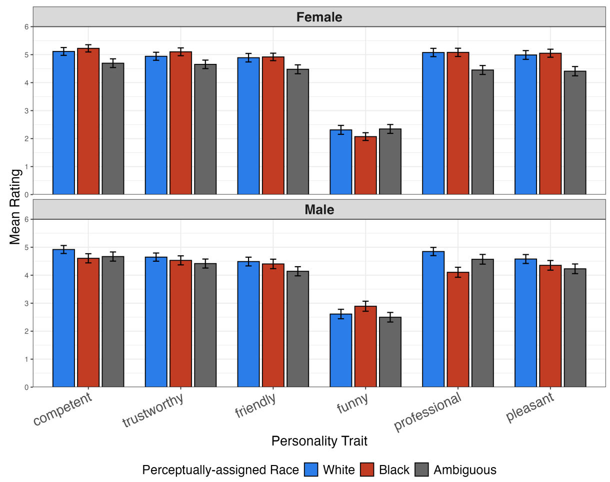

Figure 1: *Mean ratings across personality traits across assigned genders.*

For male-sounding voices, voices categorized as Black were rated as less pleasant ($\beta = -0.64, p < .05$), professional ($\beta = -1.09, p < .05$), trustworthy ($\beta = -0.53, p < .05$), and competent ($\beta = -0.67, p < .05$) than voices categorized as White; no differences were observed for friendliness or funniness. Participant ethnicity was not significant in any model.

In addition, we find that perceived human-likeness was a significant positive predictor for all personality traits across both perceived genders ($p < .01$).

3.3. Acoustic Results

For the acoustic analysis, Ambiguous voices were excluded, as the analysis focused on voices that elicited clear race categorizations. This filtering resulted in a gender imbalance across perceptually assigned race labels, with substantially fewer Black Female voices. For this reason, the Gradient Boosted Decision Tree classifier was trained only on acoustic data from male-rated voices.

A model including all acoustic features was evaluated using the CV procedure described in §2.3. To account for vowel-specific dependencies, vowel identities were one-hot encoded and included as features in the model. As a negative control, a feature consisting entirely of zeros was also included. In total, 61 features were supplied to the model. Average accuracy across test folds for the model was 65%.

Figure 2 showcases the mean gain values for the top 15 acoustic features used in a model fit to the entire training data.

A reduced model with only these top 15 features was fit using the same CV procedure. Average accuracy on test folds was 66%, validating these particular features, and indicating that a reduced feature set generalizes better to our data.

The mean centered/normalized values across normalized time for the top features CPP, F4, Residual H1*, H2*-H4*, and H2kHz* - H5kHz* are depicted in Figure 3. Given that these features are not dependent on vowel quality, mean values are aggregated across all vowels in the dataset.

What we see is that voices rated as Black have lower values for CPP, H2* - H4*, and H2kHz * - H5kHz*, as well as higher values of F4 than voices rated as White. The most robust effect is seen in Residual H1*, in which speakers rated as Black appear to have much lower values across the duration of the vowel.

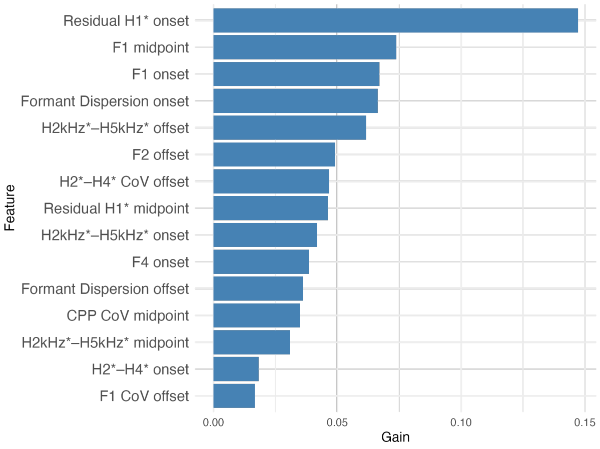

Figure 2: *Gain values for acoustic features used in the model fit to the entire data.*

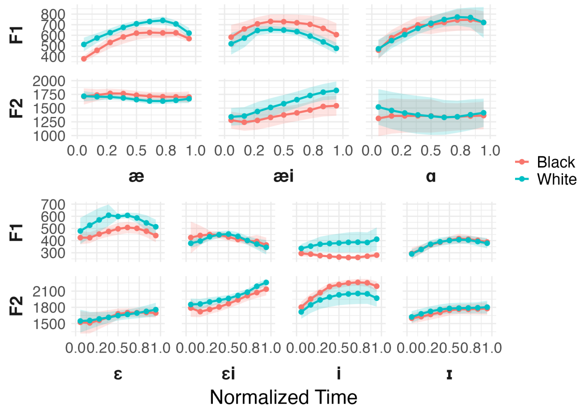

Figure 4: *Raw F1 and F2 values access time for Black versus White-rated voices. Shaded ribbons represent 95% CI around mean values.*

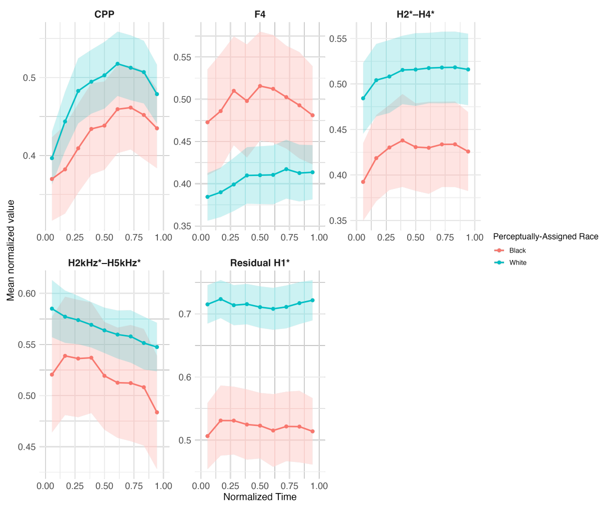

Figure 3: *Mean structure across normalized time for the top features used in the classifier. Shaded ribbons show 95% CI across mean values.*

F1/F2 values across vowels in the dataset appear in Figure 4 In Hertz (Hz). For the purpose of interpretability, the figure shows raw values. Localized differences in F1 ($\sim$100Hz) are observed for the vowels /æ/, /ɛ/, and /i/, as well as in both F1 and F2 for /æɪ/.

4. Discussion

In this study, we find that listeners are capable of categorizing synthesized voices by race, though we do find a bias towards rating voices as White, matching previous research [7]. Despite this, voices that were consistently rated as Black were also rated as significantly lower in terms of personality ratings. No such differences were found amongst female voices. One potential reason for this could be that listeners were less certain about the race of voices originally labeled as Black Females, thus making it less likely for them to assign racially-driven stereotypes. We also find that an increase in perceived human-likeness of the synthetic voice leads to an increase in all personality ratings, in line with previous research that suggests that humans are more likely to anthropomorphize synthetic speech the more human-like it sounds [10, 11].

With regards to our acoustic results, research on the phonetic differences between humans who speak MAE and AAE in the spectral domain is limited. Regardless, we know that the features we found related to lower frequency harmonics (Residual H1* and H2* - H4*) have been found to be correlated with laryngeal configuration and glottal constriction [26, 19]. The other features we find (CPP, H2kHz* - H5kHz*, and F4) are correlated with individual variation between speakers of the same language [23]. The acoustic differences we find in vowel quality (F1/F2), however, have been reliably shown to occur amongst human speakers of MAE and AAE. Overall, our results show that modern TTS models are capable of encoding fine-grained acoustic differences found amongst human MAE and AAE speakers.

Together, these findings have important implications for the implementation of Voice-AI technology. Our acoustic results can be leveraged in order to better understand how to encode the perception of race in synthetic speech. However, the presence of racial stereotyping in listener judgments warrants caution on behalf of web-developers when employing synthetic speech in socially sensitive contexts, so as to avoid perpetuating harmful racial stereotypes.

5. Conclusion

In this study, we show that listeners are able to get a sense of race when hearing synthetic speech. Though we do find an overall bias towards rating voices as White, we find that voices consistently rated as Black are generally rated as lower in terms of personality traits. In an exploratory acoustic analysis, we find that voices with lower Residual H1* and high-frequency spectral tilt are more regularly rated as Black. Features related to vowel quality (e.g., F1 and F2) also employ vowel-specific variation between voices rated as Black versus White, matching human data. Ultimately, we see that synthetic speech is capable of encoding acoustic variation humans associate with race in its signal, though more work can be done to understand not only how this is done, but also the social consequences related to it.

6. Acknowledgments

The authors would like to thank ISCA and the organizing committees of past Interspeech conferences for their help and for kindly providing the previous version of this template.

7. Generative AI Use Disclosure

Generative AI tools were solely used for code refinement and for help configuring the typesetting. Generative AI tools were not used to assist with writing the actual text of the manuscript.

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